JUAN FELIPE MALAVER

+57 301 398 2292 | juanfelipemalaver00@gmail.com | Bogotá, Colombia | LinkedIn | Github | Streamlit | Vercel |

Analytics Consultant & Data Scientist specialized in advanced quantitative modeling, applied Machine Learning, and strategic consulting. Trained at the London School of Economics and Universidad del Rosario, with a Master's in Business Analytics (fully funded by scholarship). I combine a solid foundation in econometrics and statistics with technical skills in Python, R, SQL, AWS, and the deployment of analytical solutions. Experienced leading high-impact projects at Mastercard, translating complex algorithms into profitable commercial strategies for senior banking leadership.

PROFESSIONAL EXPERIENCE

Mastercard — *Bogotá, Colombia*

Associate Analytics Consultant — *December 2025 - Present*

- Developed robust ETL pipelines in Python and SQL to automate the ingestion of large-scale transactional data, optimizing the team's analytical performance working with PII always under GDPR norms.
- Designed and implemented Machine Learning solutions (K-Medoids Clustering with Gower Distance) and optimization models to segment merchants, generating over COP 5,000 million in annual profitability following commercial deployment.
- Applied causal inference models (Propensity Score Matching and Difference-in-Differences) in Python to evaluate personalization campaigns, increasing the average spend lift by 3% per iteration.
- Translated predictive models and data science metrics into executive visualizations for bank boards of directors, facilitating data-driven strategic decisions.

Associate Consultant I. — *January 2025 - December 2025*

- Built commercial projection and predictive models in Python, R, and SQL to optimize mass campaigns in Latin American banking, facilitating data ingestion, feature engineering, data governance and data quality protocols.
- Designed segmentation and scoring methodologies based on RFM models, increasing the monetary effectiveness of loyalty programs by 23%.
- Built an ensemble model using classification algorithms (Random Forest and XGBoost) to optimize communication strategy and flag propensity of clients to buy certain products depending on their characteristics, translating the model in archetypes for the marketing team.

Universidad del Rosario — *Bogotá, Colombia*

Econometric Research Assistant — *January 2024 - January 2025*

- As an outsourced contractor, developed automations and advanced econometric models (Callaway-Sant'Anna DiD) for research projects with Princeton, Penn State, and the University of Chicago; processed geospatial databases and used Python/R for satellite image analysis.

Teaching Assistant in Statistics — *January 2023 - January 2024*

- Taught university courses with four weekly hours of lectures and two hours of office hours for Economics and Applied Mathematics to Computer Science students. The course was based on R and Python coding and covered sampling designs, bias, descriptive statistics, data visualization, probability, statistical distributions, inference, hypothesis testing, regression and modeling, panel data, DiD, PSM, and Monte Carlo.

EDUCATION

London School of Economics (LSE) — *London, United Kingdom*

Undergraduate Degree in Economics — *January 2021 - October 2025*

- Upper Second-Class Honours | Minor in Poverty and Inequality | Student Voice Group Champion, 2022-2023

Universidad del Rosario — *Bogotá, Colombia*

Undergraduate Degree in Economics — *July 2020 - October 2025*

- Minor in International Trade and Finance | GPA: 4.5/5.0 | Graduated with academic honors | Selected to deliver the graduation speech | Academic scholarship, 2020-2025
- Best final project in Applied Microeconomics in the faculty, 2021

Masters in business Analytics — *January 2026 - In progress*

- 100% academic scholarship | GPA: 4.8/5.0

PROJECTS AND INDEPENDENT PORTFOLIO

TreeStrat - Decision Tree Ensemble Business Studio (Streamlit)

- Developed an interactive Streamlit application for decision tree ensemble modeling aimed at business forecasting and scenario simulation, translating complex variable interactions into actionable business rules and revenue optimization.

ClusterLab - Adaptive Clustering Tool Business (Vercel)

- Built and deployed an interactive web tool for dynamic customer segmentation using unsupervised clustering algorithms, enabling marketing and product teams to visualize clusters in real time and maximize the ROI of commercial campaigns, specifically optimized for non-technical users and SME owners.

ML_for_ALL – Educational Repository in R for non-technical users that want to learn ML

- A GitHub repository featuring synthetic datasets designed to provide non-technical users with an introduction to machine learning, focusing on data balancing, logistic regression, classification algorithms, ensemble models for upselling, and time series analysis. It includes detailed explanations throughout the code and tips for understanding the potential business applications of each algorithm.

AWARDS AND RECOGNITIONS

Diploma of Excellence | *Ministry of Education | November 2025*

- Awarded for one of the best results in the country on the national Saber Pro exam: ranked 35th among all degree programs and 5th among economists.

Award of Excellence | *Universidad del Rosario | November 2025*

- Annual recognition for academic, cultural, athletic, social, research, and entrepreneurial commitment given to the year's most outstanding alumni.

100% Academic Scholarship for Master's Programs | *Universidad del Rosario | July 2025*

- Scholarship covering 100% of tuition for graduate programs.

Global Innovation Challenge Finalist | *Mastercard | July 2025*

- Global finalist in Mastercard's intern competition. Led the design of Innova, an AI platform that used Mastercard data to drive card payment adoption among SMEs through vendor recommendations, budgeting, customer segmentation, and personalized advice.

Onboarding Case Winners | *Mastercard | April 2025*

- Participated in a regional consulting-division competition with teams from Latin America and the Caribbean, serving as data analyst and presenter. The team placed first in Colombia and third in LAC.

Excellence Gala award - Best intern of the year | *Mastercard - Universidad del Rosario | 2025*

- Joint recognition sponsored by Universidad del Rosario and companies that offer internships for the university, awarded to the best Intern in the Year according to performance and evaluation of the firm + other criteria from the university

100th Percentile on Saber Pro | *National | July 2024*

- When taking the mandatory national exam that measures basic competencies and general and specific knowledge of future professionals in Colombia, I achieved the 100th percentile of my cohort.

Semifinalist in NCC - National Consulting Case | *National | November 2024*

- Represented Universidad del Rosario in the National Consulting Competition. My team advanced to the final stage; I provided advisory support to companies such as Pavco and solved business cases proposed by BCG.

4th Place in the National Academic Economics Competition, CANE | *National | 2024*

- I was selected to represent Universidad del Rosario for having the top GPA in the faculty in econometrics and statistics, and achieved 4th place nationally.

LANGUAGES AND SKILLS - SOFTWARE

Spanish: Native

English: C1 (IELTS Band 7.5, certified)

French: Basic knowledge (A1/A2)

Italian: Basic knowledge (A1/A2)

- **Python (Advanced):** ETL pipeline development, predictive modeling, automation, and machine learning using Scikit-Learn, Pandas, NumPy, PyTorch, PySpark, Jupyter, TensorFlow, and VS Code.
- **R (Advanced):** Advanced econometric modeling, time series, spatial analysis, and big data processing.
- **SQL (Advanced):** Complex queries for data extraction and manipulation at large scale (Hadoop, Impala, SQLite, MySQL).
- **AWS (Basic / Preparing for MLA-C01 certification):** Data preparation, design, and deployment of Machine Learning workflows and Amazon SageMaker, S3, Athena, DataWrangler, AmazonGlue.
- **Data ingestion Platforms (Databricks, Dataiku, Knime):** Data analysis, Business Intelligence, and enterprise ETL workflows.
- **Predictive Modeling and Classification:** Supervised and unsupervised algorithms, segmentation, and advanced clustering (K-Medoids, Gower Distance).
- **Causal Inference and Econometrics:** Propensity Score Matching (PSM), Difference-in-Differences (DiD, Callaway-Sant'Anna approach), and Monte Carlo simulations.
- **Streamlit and Vercel (Intermediate/Basic):** Rapid creation and deployment of interactive applications and business-oriented analytical tools (e.g., TreeStrat Business Studio and Adaptive Clustering Tool).
- **Business Intelligence (Tableau, Power BI):** Design of executive dashboards and interactive boards for C-level decision-making.
- **Office Tools:** Advanced Excel (Macros, Solver, Power Query, Power Pivot), LaTeX for technical and academic documentation.
- **AI:** MCP, Skills, Agents – agent/sub-agent teams, development via Antigravity, prompt engineering, instruction customization, vector databases, CrewAI, embeddings and fine-tuning, AI governance and risk.

MAIN CERTIFICATIONS

Leveraging Machine Learning for Business — *Wharton - 2026*

Lean Six Sigma Yellow Belt — *Universidad de los Andes - Ongoing*

MLA -C01 (Machine Learning Associate Engineer) — *AWS - Ongoing*

Note: For more details on my other certifications please refer to LinkedIn